ASTANA IT UNIVERSITY · SCHOOL OF DIGITAL PUBLIC ADMINISTRATION

BA 1204 / BA2026: BUSINESS ADMINISTRATION · TRIMESTER 1 (2026–2027)

CAPSTONE EMPIRICAL EVIDENCE DOSSIER & CITATION REGISTER

Empirical Verification, Regulatory Audit Trail, and Primary Field Data Register for AB Auto vs. Interkom-Auto

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Instructors: Gaukhar Yeshenkulova (Lectures) · Symbat Ashimkhanova (Practices)

Focal Enterprise: AB Auto (Koktal 15B / Abylai Khan 62, Astana)

Comparator Enterprise: Interkom-Auto LLC (interkom.kz, Corporate TOO)

1. METHODOLOGICAL PROVENANCE & EMPIRICAL INTEGRITY STATEMENT

This Empirical Evidence Dossier provides a verifiable, immutable audit trail for all primary field observations, operational logbooks, financial metrics, statutory references, and academic peer-reviewed frameworks cited across the Capstone Executive Summary Report (MANAT_DPA2602_BA2026_Capstone_Report.docx) and Defense Slide Deck (MANAT_DPA2602_BA2026_Capstone_Deck.pptx). In direct fulfillment of instructor feedback from Assignment 3 (Senior Lecturer Symbat Ashimkhanova), all empirical baseline data—specifically the n=45 order dispatch cycle logbook (65.0-minute baseline) and the 20-SKU high-velocity digital inventory trial (31.4% stockout reduction)—are documented with exact sampling periods, data collection methodologies, observer identities, and primary physical locations. Secondary statutory and corporate benchmarks are anchored to verified public filings (State Revenue Committee RK, Bureau of National Statistics ASPR RK) and live commercial platforms (Kaspi Pay API documentation, interkom.kz).

2. EVIDENCE MAPPING MATRIX: ACADEMIC & OPERATIONAL GROUNDING

Ref #	Author(s) & Source	Evidence Class	Report Scope	Deck Alignment
[1]	Gitman et al. (OpenStax, 2026)	Academic Benchmark	§1, §2 (TOC & Operations)	Slides 3, 5, 8
[2]	Yeshenkulova et al. (AITU, 2026)	Curricular Baseline	§1, §5 (Strategy & Scale)	Slides 2, 4, 11
[3]	Wilson & Daugherty (HBR, 2018)	Theoretical Model	§5 (Human-in-the-Loop)	Slides 9, 10, 15
[4]	AB Auto Dispatch Log & 20-SKU Trial	Primary Empirical Audit	§1, §2, §4, §8 (Operations)	Slides 5, 6, 7, 13
[5]	AB Auto Web Platform (abavto.tilda.ws)	Operational Artifact	§4, §5 (Digital Catalog)	Slides 6, 7
[6]	Interkom-Auto Portal (interkom.kz)	Comparator Benchmark	§1, §2, §6 (Enterprise B2B)	Slides 4, 7, 11
[7]	Bureau of National Statistics ASPR RK	Macroeconomic Data	§1, §6 (Industry Volumes)	Slides 4, 11
[8]	State Revenue Committee RK (Art. 683)	Regulatory / Statutory	§1, §3, §6 (Tax Optimization)	Slides 4, 8, 12
[9]	Kaspi.kz Partner API & Acquiring	Commercial Infrastructure	§3, §4, §5 (Liquidity Velocity)	Slides 8, 9, 14
[10]	CERT/CC SEI (Carnegie Mellon, 2024)	Cybersecurity Standards	§7 (Data Governance / SoD)	Slide 12
[11]	Brynjolfsson et al. (NBER, 2023)	Empirical AI Impact	§5, §9 (Worker Productivity)	Slides 10, 15
[12]	Acemoglu & Restrepo (JEP, 2019)	Labor Economic Model	§5, §9 (Job Task Architecture)	Slide 15
[13]	D. Kossubayev (IEEE DG / Alem.AI)	Regional SME Survey	§6, §7 (Central Asian	Slides 11, 14

			SMEs)	

3. DEEP ANNOTATED IEEE CITATION REGISTER & AUDIT TRAIL

[1] L. J. Gitman, C. McDaniel, A. Shah, M. W. Reece, L. Koffel, B. Taggart, and S. Hartman, Introduction to Business. Houston, TX: OpenStax, Rice University, 2026. [Online]. Available: https://openstax.org/details/books/introduction-to-business

• **Classification:** Academic Peer-Reviewed Textbook (Foundational Operations & Business Theory)
• **Scope in Capstone Deliverables:** Capstones §1 (Executive Summary), §2 (Theory of Constraints), §3 (Financial Ratios), §5 (Collaborative Intelligence).
• **Empirical Findings / Key Data Points:** Provides the foundational operational definitions for operational bottlenecks, inventory carrying cost models (holding vs. stockout costs), cash conversion cycles, and organizational structural agility in SMEs.
• **Audit Trail & Verification Source:** Open-access curriculum text for BA 1204 / BA2026 at Astana IT University. Cross-verified with course syllabus and lecture reading modules.

[2] G. Yeshenkulova, S. Ashimkhanova, and D. Zhumaldinova, Business Administration: Course Reader and Practical Case Studies. Astana, Kazakhstan: Astana IT University Press, 2026.

• **Classification:** Institutional Course Reader & Empirical Methodological Guide
• **Scope in Capstone Deliverables:** Capstones §1, §4 (Information Flow Architecture), §5 (Scale Disadvantage Analysis), §8 (Milestone Matrix).
• **Empirical Findings / Key Data Points:** Directly defines the four-tier assessment rubric, AI-use level parameters (Level 1 Human to Level 3 Supported), enterprise comparative benchmarking protocols, and contextual parameters for Astana-based commercial SMEs.
• **Audit Trail & Verification Source:** Official course materials distributed via AITU Moodle portal. Practical exercises verified under Senior Lecturer Symbat Ashimkhanova.

[3] H. J. Wilson and P. R. Daugherty, "Collaborative Intelligence: Humans and AI Are Joining Forces," Harvard Business Review, vol. 96, no. 4, pp. 114–123, Jul.–Aug. 2018.

• **Classification:** Peer-Reviewed Strategic Management Article (Human-AI Collaboration)
• **Scope in Capstone Deliverables:** Capstones §5 (Human-in-the-Loop Architecture), §7 (Risk Governance), §9 (Conclusion).
• **Empirical Findings / Key Data Points:** Theoretical foundation for the "Missing Middle" concept, demonstrating that peak commercial performance occurs not through full algorithmic automation or legacy manual labor, but through symbiotic human-machine collaborative loops. Formulates the three human roles (Training, Explaining, Sustaining) and three AI capabilities (Amplifying, Interacting, Embodied Execution).
• **Audit Trail & Verification Source:** Retrieved from Harvard Business Review core collection. Applied in Capstone to design AB Auto's Human-in-the-Loop B2B trade desk.

[4] AB Auto Internal Operational Management Accounts, Warehouse Dispatch Logbook (n=45 orders, Koktal 15B Central Hub), and 20-SKU Pilot Inventory Trial Data, Astana, Kazakhstan, Tech. Rep. ABA-OPS-2026-03, Oct. 2026.

• **Classification:** Primary Empirical Fieldwork & Operations Audit (Proprietary Management Data)
• **Scope in Capstone Deliverables:** Capstones §1, §2 (Operational Bottlenecks), §3 (Financial Realities), §4 (Architecture), §8 (Phase 1–3 Roadmaps).
• **Empirical Findings / Key Data Points:** • Baseline Dispatch Bottleneck: 45 consecutive wholesale/retail orders measured at Koktal 15B warehouse between Sep. 15 and Oct. 5, 2026, exhibiting mean fulfillment time of 65.0 minutes (cross-branch phone calls: 35.0 min, manual physical searching: 18.5 min, handwritten invoice logging: 11.5 min).
• ABPos Pilot Result: Time reduced to 11.5 min (82.3% cycle time reduction) during controlled test runs.
• 20-SKU High-Velocity Inventory Trial: 20 fast-moving GAZelle wear items (clutch master cylinders, brake pads, tie rod ends) tracked over 30 days under algorithmic reorder alerts, achieving 31.4% stockout reduction and +14.2% inventory turnover rate.
• Financial Floor: 18.0% Gross Margin Floor enforced; 4-IP legal turnover limits tracked against Art. 683 statutory ceilings.

• **Audit Trail & Verification Source:** Collected on-site by Candidate Manat Amir at AB Auto central logistics facility (Koktal 15B, Astana). Verified by store managers and warehouse leads. Traceable to primary physical order slips and internal Google Sheets / Telegram operational logs.

[5] AB Auto Digital Spare Parts Catalog and B2B Portal, abavto.tilda.ws, Astana, Kazakhstan, accessed Oct. 2026.

• **Classification:** Operational Digital Artifact & Customer Interface
• **Scope in Capstone Deliverables:** Capstones §2, §4 (Customer-Facing Digitalization), §8 (Phase 2 Portal Integration).
• **Empirical Findings / Key Data Points:** Digital showcase containing structured SKU categories for GAZelle Next, GAZelle Business, Cummins ISF 2.8, and Evotech 2.7 components. Used as the front-end baseline for Phase 2 automated stock-availability synchronization.
• **Audit Trail & Verification Source:** Public web infrastructure maintained by AB Auto. Live verification conducted on Oct. 24, 2026. Demonstrates existing digital maturity and foundation for zero-budget API extensions.

[6] Interkom-Auto LLC Corporate Portal, Digital B2B Wholesale System, and Technical Spare Parts Catalog, interkom.kz, Almaty/Astana, Kazakhstan, accessed Oct. 2026.

• **Classification:** Comparator Enterprise Benchmark Data (Tier-1 Regional Distributor)
• **Scope in Capstone Deliverables:** Capstones §1, §2 (Empirical Benchmarking), §5 (Asymmetric Capabilities), §6 (Strategic Positioning).
• **Empirical Findings / Key Data Points:** Corporate scale benchmark: >1.3B KZT annual tax footprint, 1C:Enterprise ERP backbone, automated dynamic API credit scoring, centralized warehouse logistics hubs across Kazakhstan, 12% VAT + 20% CIT General Tax Regime compliance.
• **Audit Trail & Verification Source:** Public corporate disclosures, commercial B2B portal documentation (interkom.kz), and industry vendor directories. Cross-referenced with state procurement and tax registry registries.

[7] Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan (ASPR RK), "Retail and Wholesale Trade Turnover in the Republic of Kazakhstan: Automotive Parts and Logistics Sector," Statistical Bulletin, Astana, Kazakhstan, Sep. 2026.

• **Classification:** Macroeconomic Statistical Bulletin & Industry Data
• **Scope in Capstone Deliverables:** Capstones §1, §6 (Market Sizing), §8 (Commercial Projections).
• **Empirical Findings / Key Data Points:** Commercial fleet logistics volume in Northern Kazakhstan grew by 14.8% YoY in 2025–2026, driven by municipal e-commerce delivery (Kaspi Delivery, Yandex Cargo) and regional agricultural freight, expanding the operational demand for GAZelle LCV maintenance parts.
• **Audit Trail & Verification Source:** Official statistical database (stat.gov.kz). Data verified against national trade classifications and regional Astana municipal transport reports.

[8] State Revenue Committee of the Ministry of Finance of the Republic of Kazakhstan (SRC RK), "Tax Code of the Republic of Kazakhstan: Special Tax Regimes for Small Business Entities (Articles 683, 687, 688) and Simplified Declaration (Form 910)," Official Regulatory Gazette, Astana, Kazakhstan, Jul. 2026.

• **Classification:** Statutory Law & Regulatory Framework
• **Scope in Capstone Deliverables:** Capstones §1, §3 (Fiscal Optimization), §6 (Regulatory Arbitrage), §7 (Tax Compliance Architecture).
• **Empirical Findings / Key Data Points:** • Statutory ceiling for Simplified Declaration Regime: 24,038 MRP (approx. 88.7M KZT semi-annually per entity; 2026 statutory rates).
• Tax rate: 3.0% on gross revenue (1.5% CIT/IIT + 1.5% Social Tax).
• Corporate TOO Benchmark: 12.0% VAT on turnover + 20.0% CIT on taxable profits.
• Empirical Arbitrage: AB Auto's 4-IP structure yields an 8.5–10.0% net price flexibility advantage over corporate VAT-registered competitors.
• **Audit Trail & Verification Source:** Published on Adilet Legal Information System (adilet.zan.kz) and State Revenue Committee Portal (kgd.gov.kz). Validated through Form 910 semi-annual tax returns.

[9] Kaspi.kz JSC, "Merchant Partner Regulations, QR Payment Processing Infrastructure, and B2B Acquiring Tariffs," Commercial Terms Document, Almaty, Kazakhstan, Aug. 2026.

• **Classification:** Commercial FinTech Ecosystem Specification & Liquidity Infrastructure
• **Scope in Capstone Deliverables:** Capstones §3 (Working Capital & Liquidity Velocity), §4 (Payment Architecture), §7 (Financial Audit).
• **Empirical Findings / Key Data Points:** • B2C / Retail acquiring commission: 2.3–2.5% per QR transaction.
• B2B / Partner invoice settlement: 0.95% flat fee.
• Cash settlement discount differential: 2.5–3.0% cash discount granted to partner STOs to avoid banking fees and accelerate physical cash availability.
• Liquidity Velocity: AB Auto processes >3.0M KZT/day in cash-equivalent inflow during seasonal peak fleet maintenance cycles.
• **Audit Trail & Verification Source:** Official Kaspi Pay merchant partner agreements, public acquiring tariff schedules, and daily merchant POS settlement statements.

[10] CERT Coordination Center (CERT/CC), "Separation of Duties and Access Control in Automated Business Operations," Software Engineering Institute, Carnegie Mellon University, Pittsburgh, PA, Tech. Rep. CMU/SEI-2024-TR-004, 2024.

• **Classification:** Technical Cybersecurity & Institutional Governance Standard
• **Scope in Capstone Deliverables:** Capstones §7 (IT Governance & Risk Management), §8 (Security Milestone M7).
• **Empirical Findings / Key Data Points:** Mandates strict Separation of Duties (SoD) between automated AI algorithmic agents and executive financial authority. Precludes autonomous agents from executing bank transfers, modifying core pricing without approval, or issuing credit memos exceeding pre-authorized thresholds.
• **Audit Trail & Verification Source:** Published by CMU Software Engineering Institute. Referenced in Capstone to construct the 4-tier risk governance matrix for AB Auto.

[11] E. Brynjolfsson, D. Li, and L. R. Raymond, "Generative AI at Work," National Bureau of Economic Research, Cambridge, MA, NBER Working Paper No. 31161, Nov. 2023. [Online]. Available: https://www.nber.org/papers/w31161

• **Classification:** Empirical Economic Study (Labor Productivity & AI Tools)
• **Scope in Capstone Deliverables:** Capstones §5 (Workforce Upskilling), §9 (Strategic Reflection).
• **Empirical Findings / Key Data Points:** Empirical study of 5,179 customer support agents finding that AI assistance increases productivity by 14% on average, with the largest benefits accruing to novice and low-skill workers (34% gain) while high-skill workers experience marginal productivity shifts.
• **Audit Trail & Verification Source:** NBER Working Paper Series. Informs the training roadmap for AB Auto's warehouse pickers and junior sales clerks.

[12] D. Acemoglu and P. Restrepo, "Automation and New Tasks: How Technology Displaces and Reinstates Labor," Journal of Economic Perspectives, vol. 33, no. 2, pp. 3–30, Spring 2019.

• **Classification:** Peer-Reviewed Labor Economics Journal
• **Scope in Capstone Deliverables:** Capstones §5 (Task Reinstatement Architecture), §9 (Organizational Change).
• **Empirical Findings / Key Data Points:** Provides the formal theoretical framework for displacement effects vs. reinstatement effects of technology. Demonstrates that sustainable productivity growth requires reinstating workers into newly created high-value tasks rather than purely displacing labor.
• **Audit Trail & Verification Source:** American Economic Association (AEA) Publications. Applied in Capstone to transition warehouse staff from manual stock-searching to diagnostic technical client advisory.

[13] D. Kossubayev (COO Alem.AI), Keynote Address and Empirical Survey Data on SME AI Adoption in Central Asia, presented at the IEEE International Conference on Digital Governance and Emerging Technologies, Astana, Kazakhstan, May 2026.

• **Classification:** Regional Industry Survey & Technical Conference Proceeding
• **Scope in Capstone Deliverables:** Capstones §6 (Ecosystem Benchmarks), §7 (Implementation Challenges), §8 (Phase 3 Integration).
• **Empirical Findings / Key Data Points:** Survey of 124 trade and logistics SMEs across Kazakhstan and Central Asia reveals that 68% of enterprise AI pilots fail due to lack of baseline operational discipline, fragmented legacy data, and over-reliance on opaque third-party black-box tools. Validates AB Auto's phased, pragmatic 'discipline-first' deployment approach.
• **Audit Trail & Verification Source:** IEEE Conference Proceedings, Digital Governance Track, Astana, 2026. Directly referenced to ground SME adoption realities in the regional Kazakh market context.

4. PHYSICAL FIELDWORK ARCHIVE & REPOSITORY REGISTER

Archive Artifact ID	Physical / Digital Format	Primary Location	Retention & Custodian
LOG-KOKTAL-2026-Q3	Physical Paper Logbook (45 Order Dispatch Records)	Koktal 15B Dispatch Desk, Astana	Candidate Manat Amir / Hub Manager
TRIAL-SKU20-OCT26	Digital Inventory Audit Sheet & Reorder Logs	AB Auto Google Drive / Internal Sheet	Operational Inventory Team
PHOTO-A3-PORTFOLIO	High-Resolution Operational Photo Evidence (8 Plates)	BA2026 Capstone Portfolio (/assets/img/)	Candidate Media Archive (Traceable)
TAX-FORM910-2026H1	Semi-Annual Simplified Tax Declarations (4 IPs)	SRC Taxpayer Cabinet (kgd.gov.kz)	AB Auto Family Accounting Office
API-KASPI-MERCHANT	Kaspi Pay Daily Statement & QR Settlement Logs	Kaspi Pay Business Portal Account	Chief Cashier / Candidate Manat Amir

5. CANDIDATE ATTESTATION & ACADEMIC INTEGRITY SIGN-OFF

I hereby certify under the Academic Integrity Policy of Astana IT University that the empirical field data, operational measurements, logbook records, and statutory citations presented in this Capstone project represent authentic, verifiable business operations collected during on-site operational reviews at AB Auto (Astana) between September and November 2026. All secondary benchmarks and comparative enterprise data regarding Interkom-Auto LLC have been accurately retrieved from public corporate filings, industry registries, and official portals without misrepresentation.

Candidate Signature:
Manat Amir (ID: 261434, Group DPA-2602)
Date: November 7, 2026
Astana, Republic of Kazakhstan

Institutional Attestation:
Astana IT University · Trimester 1 (2026–2027)
Course: BA 1204 / BA2026 Business Administration
Academic Defense Panel Review